ACTIVE OPTICAL DATA CABLE SYSTEM WITH INTEGRATED PROCESSING UNIT

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October 9, 2026

Title: ACTIVE OPTICAL DATA CABLE SYSTEM WITH INTEGRATED PROCESSING UNIT

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Abstract

An active data transmission system comprising a massively parallel optical interconnect with a high-performance processing unit integrated directly into its connector housing. The interconnect utilizes a two-dimensional array of photon emitters and a corresponding array of photon sensors to transmit data at rates exceeding the memory bandwidth of a host computer's central processing unit (CPU). To overcome this bottleneck, a dedicated processing unit, such as a Graphics Processing Unit (GPU), Field-Programmable Gate Array (FPGA), or Application-Specific Integrated Circuit (ASIC), is integrated into the connector head. This "active cable" is capable of performing complex data processing tasks—such as data filtering, transformation, encoding, decoding, and pixel-based network routing—on the data stream in-flight, at or near the full line rate of the optical interconnect. The system creates a new architectural paradigm where significant computational tasks are offloaded from the host system to the communication hardware itself.

Background of the Invention

Data transmission bandwidth in high-performance computing has advanced to the point where the interconnect itself is no longer the primary bottleneck. Technologies such as massively parallel optical interconnects, which transmit data using arrays of emitters and sensors, can achieve theoretical data rates in the hundreds of terabits per second (Tbps). This bandwidth now significantly exceeds the memory bandwidth of even the most powerful conventional CPUs and GPUs.

This creates a new architectural problem: a host system cannot generate or consume data fast enough to saturate the communication channel, and it cannot process the incoming data stream in real-time. This "processing bottleneck" limits the practical application of ultra-high-bandwidth interconnects, particularly in fields like quantum computing, large-scale data analysis, and real-time sensor fusion, which require both massive data throughput and intensive computation. There is a need for a system that can bridge the gap between the immense speed of modern interconnects and the processing limitations of host computers.

Summary of the Invention

The present invention is an "active" optical data cable system that solves the processing bottleneck by integrating a high-performance processing unit directly into the physical connector housing of a massively parallel optical interconnect.

The system comprises a transmitter unit with an array of photon emitters and a receiver unit with an array of photon sensors, capable of transmitting data at multi-terabit-per-second rates. The key innovation is the inclusion of a dedicated, high-performance processing unit, such as a GPU, FPGA, or ASIC, within the physical housing of the cable's connector head.

This integrated processor has direct, high-speed access to the data stream as it is being transmitted or received. It can execute a variety of computational tasks on the data in-flight, at or near the full line rate of the interconnect. These tasks can include, but are not limited to: data encoding/decoding, real-time compression/decompression, data filtering, data transformation, and complex algorithmic processing.

In one embodiment, the integrated processor performs a novel method of pixel-based network routing. It analyzes incoming data packets and encodes their destination by map-

ping them to specific emission characteristics (e.g., color) of the pixels in the emitter array, effectively turning the cable into a software-defined network switch.

The active cable system creates a new tier in the computing architecture, offloading tasks from an overburdened host CPU to the communication hardware itself. This allows the host system to work with a pre-processed, more manageable data stream, enabling the full potential of ultra-high-bandwidth interconnects to be realized.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Government and its agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental and non-commercial purposes.

Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the architecture of the active optical data cable system, showing the integrated processing unit within the connector head.

FIG. 2 is a diagram contrasting the data flow in a traditional passive cable with the data flow in the active cable of the present invention.

FIG. 3 is a conceptual diagram illustrating the method of pixel-based network routing performed by the integrated processing unit.

Detailed Description of the Invention

Referring to **FIG. 1**, the active optical data cable system (100) provides an interface between a Host Computing Device (101) and a Receiving Computing Device (102). The system is comprised of two interlocking connector heads (110a, 110b).

The transmitter (TX) connector head (110a) houses a massively parallel photon emitter array (111) (e.g., a micro-OLED display) and a high-performance Integrated Processing Unit (112) (e.g., a GPU, FPGA, or ASIC). The receiver (RX) connector head (110b) houses a

corresponding photon sensor array (113) (e.g., a CMOS image sensor) and can optionally contain its own Integrated Processing Unit (114).

Data originates from the Host (101). Instead of being directly encoded for optical transmission, it is first sent to the Integrated Processing Unit (112) within the connector head. This processor performs a pre-defined computational task on the data. This could be a data transformation, filtering, or, in a key embodiment, a routing decision. After processing, the data is passed to an encoder circuit (115), which translates the processed data into a sequence of frames for the emitter array (111).

The light frames are captured by the sensor array (113) in the RX head (110b). The captured data can either be passed to an optional integrated processor (114) for further in-flight processing (e.g., decoding, error correction) or sent directly to a decoder circuit (116), which reconstructs the digital data and passes it to the Receiving Computing Device (102).

FIG. 2 contrasts the workflow of the present invention with a conventional system. In a traditional passive interconnect (200), a Host CPU (201) is responsible for all data processing. It must perform its computation and then handle the encoding of the data for transmission via the passive cable (202). The data rate is thus limited by the Host CPU's ability to perform both tasks. In the Active Cable system (210) of the present invention, the Host CPU (211) offloads a significant computational task (e.g., "In-Flight Processing Task") to the Integrated Processing Unit (213) located inside the active cable (212). The Host CPU sends high-level data, and the active cable performs the intensive processing in-flight, delivering a final, processed result to the receiver. This frees the Host CPU to focus on other tasks and allows the system to operate at the full bandwidth of the optical interconnect.

FIG. 3 illustrates a specific application enabled by the active cable: pixel-based network routing. A data stream containing packets intended for different network destinations (e.g., Network A (301), Network B (302), Network C (303)) arrives at the Integrated Processing Unit (310) in the connector head. The processor (310) acts as a software-defined network switch. It executes a routing algorithm and assigns each data packet to a specific emission characteristic. For example, it maps all data for Network A to the red color channel, Network B to the green channel, and Network C to the blue channel. It then instructs the encoder to create a frame (320) where the pixels' colors represent the data's destination. A pixel (321) containing data for Network A is colored red, a pixel (322) for Network B is green, and a pixel (323) for Network C is blue. The receiver can then decode this frame and route the

data from each pixel based on its color, effectively multiplexing multiple network streams over a single physical link.

Claims

What is claimed is:

1. An active data transmission system, comprising:

    - a connector housing;
    - a two-dimensional array of individually addressable electromagnetic radiation emitters contained within the connector housing;
    - a high-performance processing unit, such as a GPU, FPGA, or ASIC, contained within the same connector housing;
    - an encoder circuit configured to receive processed digital data from the integrated processing unit and to generate a sequence of frames by mapping said processed data to the emission characteristics of the emitters in the array; and
    - wherein the integrated processing unit is configured to receive an initial stream of digital data from a host computer and to perform a computational task on said data in-flight before it is passed to the encoder circuit.

2. The system of claim 1, further comprising a receiver unit, the receiver unit including:

    - a second connector housing;
    - an array of electromagnetic radiation sensors contained within the second connector housing, configured to capture the sequence of frames; and
    - a decoder circuit configured to generate a replica of the processed digital data by interpreting the emission characteristics of the captured frames.

3. The system of claim 2, wherein the receiver unit further comprises a second high-performance processing unit contained within the second connector housing, configured to perform a computational task on the data after it is captured by the sensor array.

4. The system of claim 1, wherein the computational task performed by the integrated processing unit is selected from the group consisting of: data filtering, data transformation, data compression, data decompression, data encoding, data decoding, and network routing.

5. The system of claim 1, wherein the data transmission rate of the system exceeds the memory bandwidth of the host computer.

6. The system of claim 1, wherein the integrated processing unit is configured to perform a pixel-based network routing task, wherein said task comprises assigning data packets to specific emission characteristics based on their network destination.

7. The system of claim 6, wherein the emission characteristic used for routing is the wavelength or color of the emitted radiation.

8. A method for active data transmission, comprising the steps of:

    - receiving, at a connector housing of a data cable, a stream of digital data from a host computer;
    - processing, by a high-performance processing unit located inside the connector housing, said stream of digital data to generate a processed data stream;
    - encoding, within the same connector housing, the processed data stream into a sequence of frames, wherein each frame comprises a plurality of emission channels corresponding to a two-dimensional array of emitters; and
    - emitting said sequence of frames from the two-dimensional array of emitters.

9. The method of claim 8, wherein the step of processing the data comprises executing a pixel-based network routing algorithm to determine the emission characteristics for each channel based on a network destination.

10. The method of claim 8, wherein the data is transmitted at a rate that is higher than the rate at which the host computer can load data from its main memory.

Drawings

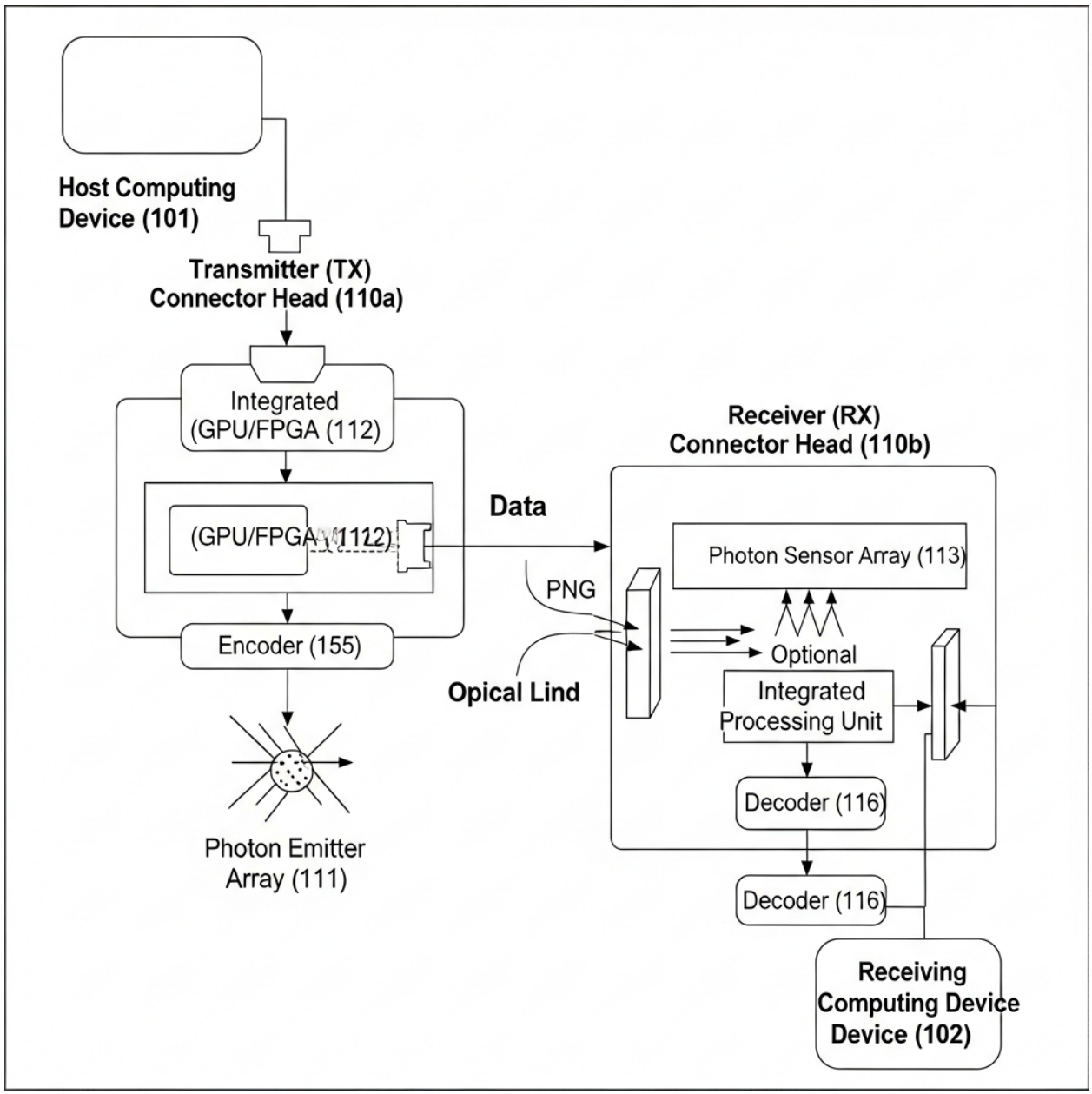
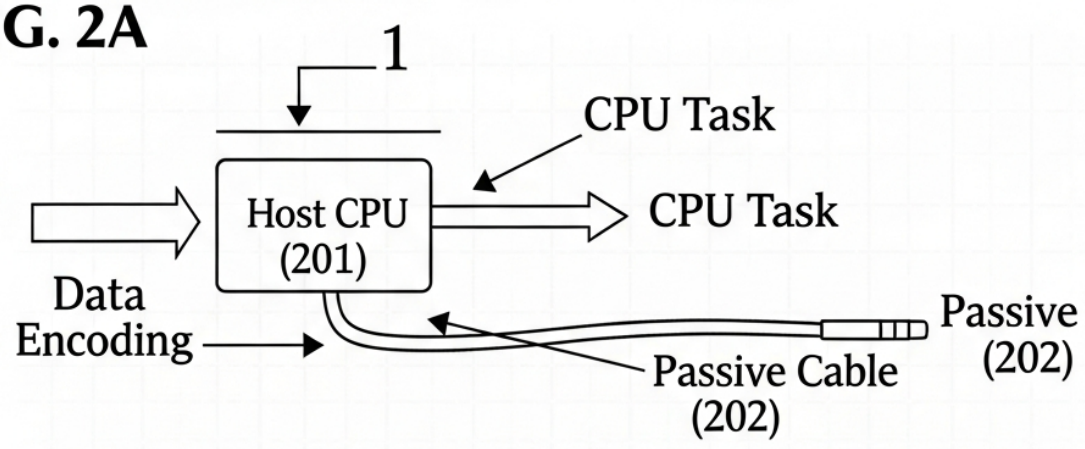


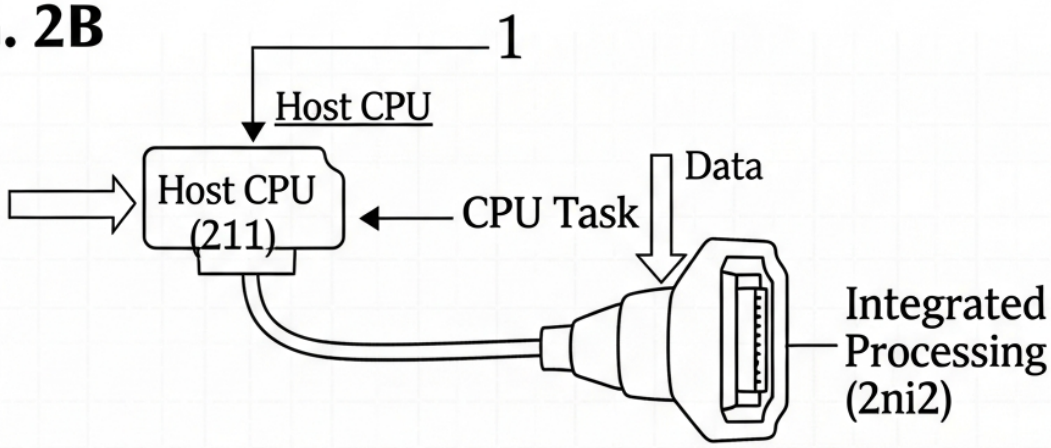
Figure 1: A block diagram illustrating the architecture of the active optical data cable system, showing the integrated processing unit within the connector head.

FIG. 2A



Traditional Passive Interconnect (200)

FIG. 2B



Active Acable System (210)

Figure 2: A diagram contrasting the data flow in a traditional passive cable with the data flow in the active cable of the present invention.

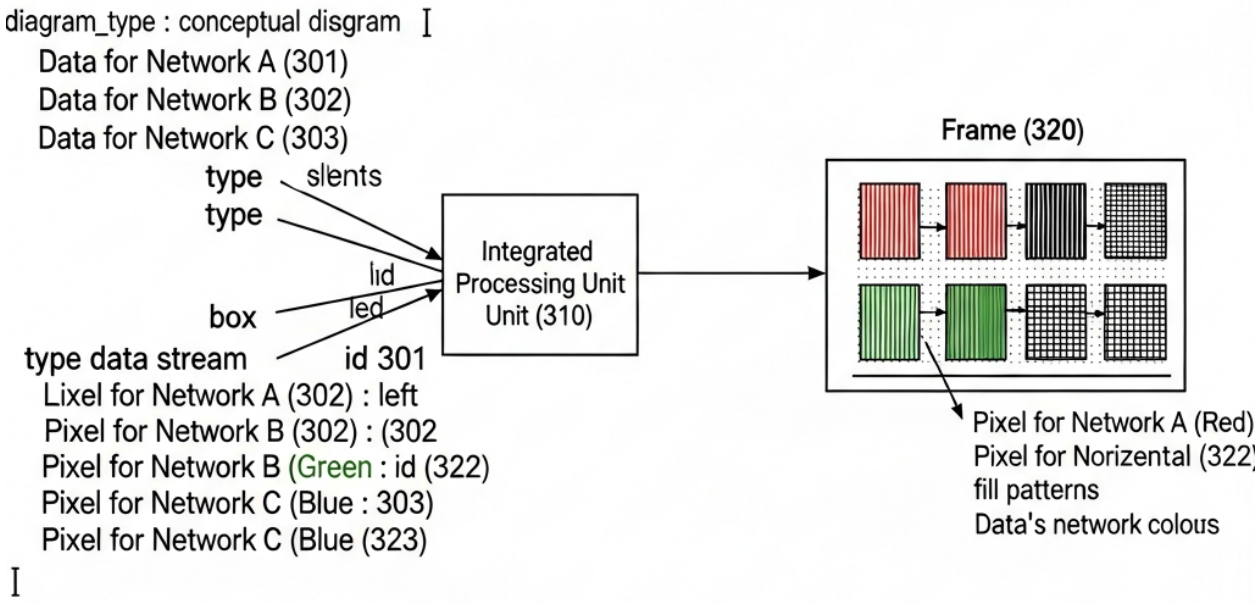


Figure 3: A conceptual diagram illustrating the method of pixel-based network routing performed by the integrated processing unit.